

Linear Algebra—Winter/Spring 2026

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Problem Set 10

(Last Edited: Tuesday, April 14, 2026)

Due: 10 PM, Tuesday April 14, 2026

Instructions

- a. *Timeline.* In general, expect the first “half” of a problem set to be released by Tuesday nights (or whenever the first lecture of the week takes place). Expect the full problem set to be released by Friday nights (or whenever the second lecture of the week takes place). The full assignment will, *typically*, be due on Tuesday nights, 8 PM.
- b. *Punctuality.* Aim to get the assignment done three days prior to the deadline. We will not be able to entertain any requests for extensions. If you seek clarification from us and we do not respond on time, do the following: (a) assume whatever seems most reasonable, and solve the question under that assumption and (b) ensure you have evidence for making the clarification request and report the incident to one of the PhD TAs.
- c. *Resources.* Feel free to use any resource you like (including generative AI tools) to get help but please make sure you understand what you finally write and submit. The TAs may ask you to explain the reasoning behind your response if something appears suspicious.
- d. *Graded vs Practice Questions.* Problems numbered using Arabic numerals (i.e. 1, 2, 3, ...) constitute assignment problems that will be graded. Problems numbered using Roman numerals (i.e. i, ii, iii, ...) are practice problems that will not be graded.

The text here is Linear Algebra Done Right, by Sheldon Axler (3rd edition); we are on Chapter 6.

No more graded problems will be added to this problem set.

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1. (This question is based on the proof of Statement 6.37 in the text)
Suppose $T \in \mathcal{L}(V)$. Prove that if T has an upper-triangular matrix with respect to some basis $v_1, \dots, v_n$ of V , then T has an upper-triangular matrix with respect to the orthonormal vectors $e_1 \dots e_n$ obtained by the Gram Schmidt procedure, i.e. where $e_1 := v_1$ and for $1 < j \leq n$,

$$e_j := \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}}{c_j}$$

where c_j is a normalisation factor that ensures $\|e_j\|^2 = 1$.

2. Suppose V is a complex inner product space. Prove that

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2 + \|u + iv\|^2 i - \|u - iv\|^2 i}{4}$$

for all $u, v \in V$.

3. On $\mathcal{P}_2(\mathbb{R})$, consider the inner product given by

$$\langle u, v \rangle = \int_0^1 p(x)q(x) \, dx.$$

Apply the Gram-Schmidt procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

- i. What goes wrong if we define $\langle v, w \rangle$ to be $\sum_{i=1}^n \alpha_i \beta_i$ where $v = (\alpha_1, \alpha_2, \dots, \alpha_n)$ and $w = (\beta_1, \beta_2, \dots, \beta_n)$ are vectors in $\mathbb{C}^n$? More concretely, is $\langle v, w \rangle$ an inner product? If so, show that all conditions for being an inner product are satisfied. If not, show which condition fails (suffices to show one, if there are many conditions that fail).
- ii. Suppose $T \in \mathcal{L}(\mathbb{R}^3)$ has an upper-triangular matrix with respect to the basis

$$\{(1, 0, 0), (1, 1, 1), (1, 1, 2)\}.$$

Find an orthonormal basis of $\mathbb{R}^3$ with respect to which T has an upper-triangular matrix.

- iii. Prove that

$$(a + b + c + d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \geq 16$$

for all positive real numbers a, b, c, d .

- iv. Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that

$$\int_0^1 p(x) \cos \pi x \, dx = \int_0^1 p(x)q(x) \, dx$$

for every $p \in \mathcal{P}_2(\mathbb{R})$.